Functional Analysis

Lecture notes

(based on the course of Prof. Robert König)

Zayn

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1 Metric and topological spaces

1.1 Metric spaces

Definition 1.1.1: A *metric space* (X, d) is a set X together with a *metric* or *distance function* $d : X \times X \to [0, \infty)$ satisfying

$$\begin{aligned} &d(x, y) \geq 0 \quad \text{with equality if and only if } x = y, \\ &d(x, y) = d(y, x), \\ &d(x, z) \leq d(x, y) + d(y, z) \end{aligned}$$

for all $x, y, z \in X$. The *open ball* of radius $r > 0$ around $x \in X$ is the set

$$B_r(x) := \{y \in X \mid d(x, y) < r\}.$$

We also define

$$\overline{B}_r(x) := \{y \in X \mid d(x, y) \leq r\}.$$

Remark 1.1.2: Note that $\overline{B}_r(x) \neq \overline{B_r(x)}$ in general (e.g., for the discrete metric), where $\overline{B_r(x)}$ denotes the closure of $B_r(x)$ (see below).

Example 1.1.3: $X := \mathbb{R}$ and $d(x, y) := |x - y|$ for $x, y \in X$. More generally, if X is a vector space and $\| \cdot \|$ a norm on X, then $d(x, y) := \|x - y\|$ defines a metric on X. For example, the set $C([a, b])$ of continuous functions on $[a, b]$ is a metric space with metric induced by the norm $\|f\| := \sup_{t \in [a,b]} |f(t)|$.

Example 1.1.4: Suppose X is an arbitrary set and $d(x, y) := 1 - \delta_{x,y}$, where

$$\delta_{x,y} := \begin{cases} 1 & \text{if } x = y, \\ 0 & \text{otherwise} \end{cases}$$

is the Kronecker delta. This is called the *discrete metric*.

Open sets in a metric space (X, d) are often defined as follows.

Definition 1.1.5: A subset $U \subset X$ of a metric space X is *open* if and only if for every $x \in U$, there is some $r > 0$ such that $B_r(x) \subset U$.

Remark 1.1.6: Definition 1.1.5 is equivalent to the statement that $U \subset X$ is open if and only if U is the union of open balls $B_r(x)$ with $x \in U$ and $r > 0$.

Remark 1.1.7: If $r > 0$ and $x \in X$, then $B_r(x)$ is open. Indeed, let $y \in B_r(x)$ be arbitrary. Then $d(y, x) < r$, and, in particular, $\varepsilon := r - d(y, x) > 0$. It follows from the triangle inequality that $B_\varepsilon(y) \subset B_r(x)$.

Lemma 1.1.8: Let X be a metric space. Let $\mathcal{T} \subset 2^X$ be the set of open subsets of X. Then the following holds:

(i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.

(ii) $\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}$ for any family $\{U_\alpha\}_{\alpha \in I} \subset \mathcal{T}$.

(iii) $\bigcap_{\alpha \in I} U_\alpha \in \mathcal{T}$ for any family $\{U_\alpha\}_{\alpha \in I} \subset \mathcal{T}$ with $|I| < \infty$.

Proof. Exercise. □

1.2 Topological spaces

Definition 1.2.1: A *topological space* $(X, \mathcal{T})$ is a set X together with a set $\mathcal{T} \subset 2^X$ of subsets (called *open sets*) satisfying

(i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.

(ii) $\bigcup_{\alpha\in I} U_\alpha \in \mathcal{T}$ for any family $\{U_\alpha\}_{\alpha\in I} \subset \mathcal{T}$.

(iii) $\bigcap_{\alpha\in I} U_\alpha \in \mathcal{T}$ for any family $\{U_\alpha\}_{\alpha\in I} \subset \mathcal{T}$ with $|I| < \infty$.

The family $\mathcal{T}$ is called a *topology* on X. A subset $A \subset X$ of a topological space X is called *closed* if it is the complement of an open set, i.e., if $A^c := X \setminus A = \{x \in X \mid x \notin A\} \in \mathcal{T}$.

Example 1.2.2: Let X be a set. Then $\mathcal{T} := \{U \subset X \mid U = \emptyset \text{ or } |U^c| < \infty\}$ is called the *cofinite topology* on X.

Example 1.2.3: If (X, d) is a metric space, then the topology given in Definition 1.1.5 is called the *topology induced by the metric* d (cf. Lemma 1.1.8). A topology induced by a metric is called a *metrizable* topology.

Remark 1.2.4: Note that the metric inducing a metrizable topology is non-unique. Two metrics inducing the same topology are called *equivalent*.

Example 1.2.5: Let X be a set of cardinality $|X| \geq 2$. Then the *trivial topology* $\mathcal{T} := \{\emptyset, X\}$ on X is not metrizable. Indeed, assume that the topology is induced by a metric d. Let $x, y \in X$ with $x \neq y$ be arbitrary and set $r := d(x, y) > 0$. Then $B_{r/2}(x)$ is an open set (with respect to the topology induced by d) which contains x but does not contain y. However, the topology $\mathcal{T}$ has no such neighborhoods.

The set of closed subsets of X satisfies the following:

Lemma 1.2.6: Let $(X, \mathcal{T})$ be a topological space. Then

(i) $\emptyset$ and X are closed.

(ii) Any finite union $\bigcup_{\alpha\in I} A_\alpha$ of closed subsets $\{A_\alpha\}_{\alpha\in I}$, $|I| < \infty$, is closed.

(iii) Arbitrary intersections $\bigcap_{\alpha\in I} A_\alpha$ of closed subsets $\{A_\alpha\}_{\alpha\in I}$ are closed.

Proof. Follows immediately from the definition of closed/open sets. □

Definition 1.2.7: Let X be a topological space and $\{U_\alpha\}_{\alpha\in I}$ a family of open subsets. Then $\{U_\alpha\}_{\alpha\in I}$ is called a *basis* of the topology if every open $U \subset X$ is a union $U = \bigcup_{\alpha\in J} U_\alpha$ of subsets belonging to the family.

Example 1.2.8: For a metric space (X, d), the set $\{B_r(x)\}_{r>0,\, x\in X}$ of all open balls is a basis of the topology, see Remark 1.1.6.

Definition 1.2.9: Let $(X, \mathcal{T})$ be a topological space. For $M \subset X$, define

$$\text{the } \textit{closure } \overline{M} \text{ of } M \text{ as } \overline{M} := \bigcap_{\substack{A \text{ closed} \\ M\subset A}} A, \qquad \text{the } \textit{interior } \operatorname{int}(M) \text{ of } M \text{ as } \operatorname{int}(M) := \bigcup_{\substack{U \text{ open} \\ U\subset M}} U.$$

Remark 1.2.10: Clearly $\overline{M}$ is closed and $M \subset \overline{M}$. In fact, $\overline{M}$ is the smallest closed set which contains M. Similarly, $\operatorname{int}(M)$ is the largest open subset contained in M.

Example 1.2.11: Let (X, d) be a metric space. It is straightforward to check that $\overline{B}_r(x)$ is closed for $r > 0$ and $x \in X$. But in general, $\overline{B_1(x)} \neq \overline{B}_1(x)$ (exercise).

Definition 1.2.12: Let $(X, \mathcal{T})$ be a topological space. A *neighborhood* N_x of an element $x \in X$ is a subset $N_x \subset X$ such that $x \in U \subset N_x$ for an open set $U \in \mathcal{T}$. In other words, N_x is a neighborhood of x if $x \in \operatorname{int}(N_x)$.

Remark 1.2.13: In a metric space this means that there is some $r > 0$ such that $B_r(x) \subset N_x$.

1.3 Induced (or relative) metric/topology

Definition 1.3.1: Let (X, d) be a metric space and $Y \subset X$. Then the restriction of the metric $d(\cdot, \cdot)$ to $Y \times Y$ is called the *induced* or *relative metric* on Y.

Lemma 1.3.2: Let X be a metric space and $Y \subset X$. A subset $U \subset Y$ is open with respect to the induced metric if and only if $U = \hat{U} \cap Y$ for some $\hat{U}$ which is open in X.

Proof. Let U be a set of the form $U = \hat{U} \cap Y$ with $\hat{U}$ open in X. Then for $x \in U$, there is some $r > 0$ such that $B_r(x) \subset \hat{U}$ because $\hat{U}$ is open. But then $B_r(x) \cap Y \subset U$, hence U is open in Y.

Conversely, suppose that $U \subset Y$ is open. Let $x \in U$. Choose $r_x > 0$ such that $B^Y_{r_x}(x) \subset U$, where

$$B^Y_r(x) := \{y \in Y \mid d(y, x) < r\} = B_r(x) \cap Y$$

for $r > 0$. Define $\hat{U} := \bigcup_{x \in U} B_{r_x}(x)$. Then $\hat{U}$ is open by definition, $U \subset \hat{U}$, and $U \subset \hat{U} \cap Y$. But we also have

$$\hat{U} \cap Y = \bigcup_{x \in U} \big(B_{r_x}(x) \cap Y\big) = \bigcup_{x \in U} B^Y_{r_x}(x) \subset U,$$

which shows that $\hat{U} \cap Y = U$. □

Definition 1.3.3: Let $(X, \mathcal{T})$ be a topological space and $Y \subset X$. Define the *induced* (or *relative*) *topology* on Y by the family of open sets

$$\{\hat{U} \cap Y \mid \hat{U} \in \mathcal{T}\}. \tag{1}$$

It is straightforward to check that (1) indeed defines a topology on Y.

1.4 Product spaces and product topology

Definition 1.4.1: Let (X_1, d_1) and (X_2, d_2) be metric spaces. Then

$$d\big((x_1, x_2), (y_1, y_2)\big) := d_1(x_1, y_1) + d_2(x_2, y_2)$$

defines a metric on $X_1 \times X_2$. The induced topology is called the *product topology.*

Remark 1.4.2: Note that there are several alternative ways of defining a metric on $X_1 \times X_2$, as for example

$$\tilde{d}\big((x_1, x_2), (y_1, y_2)\big) := \max\{d_1(x_1, y_1),\, d_2(x_2, y_2)\},$$
$$\hat{d}\big((x_1, x_2), (y_1, y_2)\big) := \sqrt{d_1(x_1, y_1)^2 + d_2(x_2, y_2)^2}.$$

Lemma 1.4.3: Let (X_1, d_1) and (X_2, d_2) be metric spaces. Then $U \subset X_1 \times X_2$ is open if and only if for every $(x_1, x_2) \in U$, there are open subsets $U_1 \subset X_1$ and $U_2 \subset X_2$ such that $(x_1, x_2) \in U_1 \times U_2$ and $U_1 \times U_2 \subset U$.

Proof. Exercise. □

Definition 1.4.4: Let X_1, X_2 be topological spaces. Then the *product topology* on $X_1 \times X_2$ is defined as follows. A set $U \subset X_1 \times X_2$ is open if and only if for every $(x_1, x_2) \in U$, there are open subsets $U_1 \subset X_1$ and $U_2 \subset X_2$ such that $(x_1, x_2) \in U_1 \times U_2 \subset U$.

Remark 1.4.5: Definition 1.4.4 implies that the family of sets of the form $U_1 \times U_2$, where $U_j \subset X_j$ is open for $j = 1, 2$, is a basis of the product topology on $X_1 \times X_2$.

This can be generalized as follows:

Definition 1.4.6 (Product set and product topology): Let $\{X_\alpha\}_{\alpha\in A}$ be a family of topological spaces (indexed by a set A). Then the *product set* $\prod_{\alpha\in A} X_\alpha$ is the family of all maps

$$x : A \to \bigcup_{\alpha\in A} X_\alpha, \qquad \alpha \mapsto x_\alpha \in X_\alpha.$$

We write $x = (x_\alpha)_{\alpha\in A}$ for elements $x \in \prod_{\alpha\in A} X_\alpha$.

The *product* (or *Tikhonov*) *topology* on $\prod_{\alpha\in A} X_\alpha$ is defined by the following basis: Every basis element $G = G(\{\alpha_j\}_{j=1}^n, \{U_j\}_{j=1}^n)$ is specified by a finite subset $\{\alpha_1, \dots, \alpha_n\} \subset A$, and an open set $U_j \subset X_{\alpha_j}$ for $j = 1, \dots, n$ as follows:

$$G = \Big\{ x = (x_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} X_\alpha \;\Big|\; x_{\alpha_j} \in U_j \text{ for all } j = 1, \dots, n \Big\}.$$

In other words, $G = \prod_{\alpha\in A} G_\alpha$, where $G_\alpha \subset X_\alpha$ is open for every $\alpha \in A$, and $G_\alpha = X_\alpha$ for all but finitely many $\alpha \in A$.

Remark 1.4.7: The *axiom of choice* states that if $X_\alpha \neq \emptyset$ for all $\alpha \in A$, then $\prod_{\alpha\in A} X_\alpha \neq \emptyset$.

1.5 Sequences in topological spaces

Definition 1.5.1: Let (X, d) be a metric space. A sequence $(x_n)_{n\in\mathbb{N}} \subset X$ *converges* to $x \in X$ if for every $\varepsilon > 0$, there is some $n_0 \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon$ for all $n \geq n_0$. We write $x_n \to x$ for $n \to \infty$.

Lemma 1.5.2 (Convergence of sequences in metric spaces): In a metric space (X, d), the following are equivalent:

(i) $x_n \to x$ for $n \to \infty$.

(ii) For every neighborhood U of x there is some $n_0 \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq n_0$, i.e., $(x_n)_{n\in\mathbb{N}}$ *eventually is in* U.

Proof. Exercise. □

Definition 1.5.3: Let $(X, \mathcal{T})$ be a topological space, $(x_n)_{n\in\mathbb{N}} \subset X$. Then $(x_n)_{n\in\mathbb{N}}$ *converges* to $x \in X$ if for every neighborhood N of x, $(x_n)_{n\in\mathbb{N}}$ eventually is in N.

Remark 1.5.4: Note that (unlike in a metric space) the limit x need not be unique in a general topological space: consider e.g., the trivial topology.

Definition 1.5.5: Let $(X, \mathcal{T})$ be a topological space, $(x_n)_{n\in\mathbb{N}} \subset X$, and $x \in X$. Then x is an *accumulation point* of $(x_n)_{n\in\mathbb{N}}$ if for every neighborhood N of x and every $n_0 \in \mathbb{N}$, there is some $n \geq n_0$ such that $x_n \in N$. We say that $(x_n)_{n\in\mathbb{N}}$ *often lies in* N, or equivalently that $x_n \in N$ for infinitely many $n \in \mathbb{N}$.

Remark 1.5.6: If $x_n \to x$ for $n \to \infty$ or $x_{n_j} \to x$ for $j \to \infty$ (i.e., for a subsequence), then x is an accumulation point of $(x_n)_{n\in\mathbb{N}}$.

Lemma 1.5.7: Suppose (X, d) is a metric space, $(x_n)_{n\in\mathbb{N}} \subset X$, and $x \in X$ an accumulation point of $(x_n)_{n\in\mathbb{N}}$. Then there is a subsequence $(x_{n_j})_{j\in\mathbb{N}}$ such that $x_{n_j} \to x$ for $j \to \infty$.

Proof. Recursively choose $n_1 < n_2 < n_3 < \dots$ such that $d(x_{n_j}, x) < 1/j$. Then clearly $x_{n_j} \to x$ for $j \to \infty$. □

Lemma 1.5.8: Let (X, d) be a metric space and $M \subset X$. Then the following are equivalent:

(i) $x \in \overline{M}$.

(ii) There is a sequence $(x_n)_{n\in\mathbb{N}} \subset M$ such that $x_n \to x$ for $n \to \infty$.

Proof. Exercise. □

Remark 1.5.9: The statement of Lemma 1.5.7 does not hold in general topological spaces. The key difference is that in metric spaces, it is sufficient to consider the countable family $\{B_{1/j}(x)\}_{j\in\mathbb{N}}$ of neighborhoods of x. Similarly, the statement of Lemma 1.5.8 does not hold in general topological spaces (exercise).

1.6 Complete metric spaces

Definition 1.6.1: Let (X, d) be a metric space. A sequence $(x_n)_{n\in\mathbb{N}} \subset X$ is called a *Cauchy sequence* if for every $\varepsilon > 0$ there is some $n_0 \in \mathbb{N}$ such that $d(x_n, x_m) \leq \varepsilon$ for all $n, m \geq n_0$. The space (X, d) is called *complete* if all Cauchy sequences converge.

Remark 1.6.2: Completeness is not a topological property! Equivalent metrics define the same convergent sequences, but not the same Cauchy sequences.

Theorem 1.6.1: Completeness of subsets of a complete space

Let (X, d) be a complete metric space and $M \subset X$. Then the following are equivalent:

(i) (M, d) is complete (with d the restriction of d to M).

(ii) M is closed.

Proof. Both directions are immediate consequences of the characterization of closed sets given by Lemma 1.5.8.

(i)⇒(ii): Suppose $(x_n)_{n\in\mathbb{N}} \subset M$ and $x \in X$ are such that $x_n \to x$ for $n \to \infty$. Convergence implies that $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in M, thus $x \in M$ by completeness of M. Hence M is closed by Lemma 1.5.8.

(ii)⇒(i): Let $(x_n)_{n\in\mathbb{N}} \subset M$ be a Cauchy sequence. By completeness of X, there is some $x \in X$ such that $x_n \to x$ for $n \to \infty$. Because M is closed, we have $x \in M$ by Lemma 1.5.8. Thus M is complete. □

Definition 1.6.3: Let (X, d) and (X', d') be two metric spaces and $f : X \to X'$. Then f is called an *isometry* if and only if

$$d'\big(f(x), f(y)\big) = d(x, y) \qquad \text{for all } x, y \in X.$$

Theorem 1.6.2: Completion of metric spaces

Let (X, d) be a metric space. Then there is a metric space $(\tilde{X}, \tilde{d})$ and a map $\iota : X \to \tilde{X}$ such that

(i) $(\tilde{X}, \tilde{d})$ is a complete metric space,

(ii) the map ι is injective and an isometry,

(iii) we have $\overline{\iota(X)} = \tilde{X}$.

Proof idea. Let $\tilde{X}$ be the set of equivalence classes of Cauchy sequences with the equivalence relation

$$(x_n)_{n\in\mathbb{N}} \sim (y_n)_{n\in\mathbb{N}} \quad :\Longleftrightarrow \quad \lim_{n\to\infty} d(x_n, y_n) = 0.$$

It can be shown that $\tilde{d}([x],[y]) := \lim_{n\to\infty} d(x_n, y_n)$ is well-defined and defines a metric on $\tilde{X}$. Also, the space $(\tilde{X}, \tilde{d})$ can be shown to be complete.

For $x \in X$, define $\iota(x) \in \tilde{X}$ as the equivalence class of the sequence $x_n = x$ for all $n \in \mathbb{N}$. It is clear that ι is injective and an isometry. Density of $\iota(X)$ in $\tilde{X}$ follows because, for every Cauchy sequence $(x_n)_{n\in\mathbb{N}}$ in X, we have

$$\tilde{d}\big(\iota(x_n), [(x_m)_{m\in\mathbb{N}}]\big) = \lim_{m\to\infty} d(x_n, x_m) \longrightarrow 0 \quad \text{as } n \to \infty,$$

by the Cauchy property. □

1.7 Compactness

Definition 1.7.1: A topological space X is called *compact* if every open cover of X contains a finite cover. A subset $K \subset X$ is called *compact* if it is compact with respect to the induced topology.

Remark 1.7.2: It is easy to see that $K \subset X$ is compact if and only if every cover $K \subset \bigcup_{\alpha\in I} U_\alpha$ with sets $\{U_\alpha\}_{\alpha\in I}$ which are open in X contains a finite subcover.

Lemma 1.7.3 (Compactness and finite intersection property): Let X be a topological space. Then the following are equivalent:

(i) X is compact.

(ii) For any family $\{A_\alpha\}_{\alpha\in I}$ of closed subsets of X such that $\bigcap_{\alpha\in I'} A_\alpha \neq \emptyset$ for any finite subset $I' \subset I$, we have $\bigcap_{\alpha\in I} A_\alpha \neq \emptyset$.

Proof. (i)⇒(ii): Suppose $\{A_\alpha\}_{\alpha\in I}$ is a family of closed subsets of X such that $\bigcap_{\alpha\in I'} A_\alpha \neq \emptyset$ for any finite subset $I' \subset I$. Assume for the sake of contradiction that $\bigcap_{\alpha\in I} A_\alpha = \emptyset$. Define $U_\alpha := X \setminus A_\alpha$ for $\alpha \in I$. Then $\{U_\alpha\}_{\alpha\in I} \subset \mathcal{T}$ is a family of open sets and

$$\bigcup_{\alpha\in I} U_\alpha = \bigcup_{\alpha\in I} X \setminus A_\alpha = X \setminus \bigcap_{\alpha\in I} A_\alpha = X$$

by assumption. That is, $\{U_\alpha\}_{\alpha\in I}$ is an open cover of X, and by compactness, there is a finite subcover $\{U_\alpha\}_{\alpha\in I'}$, $I' \subset I$, $|I'| < \infty$. But then

$$\emptyset = X^c = X \setminus \bigcup_{\alpha\in I'} U_\alpha = \bigcap_{\alpha\in I'} X \setminus U_\alpha = \bigcap_{\alpha\in I'} A_\alpha,$$

a contradiction.

(ii)⇒(i): Let $X = \bigcup_{\alpha\in I} U_\alpha$ be a cover of X by open sets $\{U_\alpha\}_{\alpha\in I}$. Define $A_\alpha := X \setminus U_\alpha$ for $\alpha \in I$. Then

$$\bigcap_{\alpha\in I} A_\alpha = \bigcap_{\alpha\in I} X \setminus U_\alpha = X \setminus \Big(\bigcup_{\alpha\in I} U_\alpha\Big) = \emptyset.$$

By assumption, this implies that there is a finite subset $I' \subset I$ such that $\bigcap_{\alpha\in I'} A_\alpha = \emptyset$. But $\bigcap_{\alpha\in I'} A_\alpha = X \setminus \big(\bigcup_{\alpha\in I'} U_\alpha\big)$, hence $\{U_\alpha\}_{\alpha\in I'}$ defines a finite subcover of X. □

Definition 1.7.4: A metric space (X, d) is called *totally bounded* if for every $r > 0$, there is a finite cover of X by sets of the form $B_r(x)$ with $x \in X$.

Theorem 1.7.1: Compactness in metric spaces (Heine–Borel)

Let (X, d) be a metric space. Then the following are equivalent:

(i) X is compact.

(ii) Each sequence in X has an accumulation point.

(iii) X is complete and totally bounded.

Proof. See Analysis I/II. □

Remark 1.7.5: If X_1, X_2 are compact metric spaces, then $X_1 \times X_2$ is compact (exercise). We will prove a more general statement for topological spaces below (Tikhonov's theorem).

1.8 Separability

Definition 1.8.1: Let X be a topological space and $M \subset X$. Then M is *dense* in X if $\overline{M} = X$. The space X is called *separable* if there is a countable dense subset $M \subset X$.

Example 1.8.2: The metric space $\mathbb{R}^n$ with $d(x, y) = \sum_{j=1}^n |x_j - y_j|$ is separable: the set $\mathbb{Q}^n$ is a countable dense subset. The same holds, e.g., for the Euclidean metric $d_2(x, y) = \left(\sum_{j=1}^n |x_j - y_j|^2\right)^{1/2}$.

Lemma 1.8.3: The normed space

$$\ell^2(\mathbb{R}) := \left\{ x = (x_j)_{j\in\mathbb{N}} \subset \mathbb{R} \;\middle|\; \sum_{j=1}^{\infty} |x_j|^2 < \infty \right\}$$

with norm $\|x\|_2 = \left(\sum_{j=1}^{\infty} |x_j|^2\right)^{1/2}$ is separable.

Proof sketch. For $n \in \mathbb{N}$, let $R_n \subset \ell^2(\mathbb{R})$ be the set of sequences $x = (x_j)_{j\in\mathbb{N}}$ satisfying $x_j = 0$ for all $j > n$. By Example 1.8.2, each R_n contains a countable dense subset $Q_n \subset R_n$. One then argues using the triangle inequality that $Q = \bigcup_{n\in\mathbb{N}} Q_n$ is a countable dense subset of $\ell^2(\mathbb{R})$. □

Lemma 1.8.4: The normed space

$$\ell^\infty(\mathbb{R}) := \left\{ x = (x_j)_{j\in\mathbb{N}} \subset \mathbb{R} \;\middle|\; \sup_{j\in\mathbb{N}} |x_j| < \infty \right\}$$

with norm $\|x\|_\infty = \sup_{j\in\mathbb{N}} |x_j|$ is not separable.

Proof. For $M \subset \mathbb{N}$ define $x^{(M)} \in \ell^\infty(\mathbb{R})$ by

$$x_j^{(M)} := \begin{cases} 1 & \text{if } j \in M, \\ 0 & \text{otherwise.} \end{cases}$$

Then $\|x^{(M)} - x^{(M')}\|_\infty = 1$ for $M \neq M'$. Thus the balls $\{B_{1/2}(x^{(M)})\}_{M\subset\mathbb{N}}$ are disjoint and open, and this family of sets is uncountable.

Assume $Z \subset \ell^\infty(\mathbb{R})$ is a countable dense subset. Then there is a point $z^{(M)} \in B_{1/2}(x^{(M)}) \cap Z$ for every $M \subset \mathbb{N}$. In particular, this gives rise to an injective map $\iota : 2^{\mathbb{N}} \to Z$, $\iota(M) = z^{(M)}$, contradicting the fact that $2^{\mathbb{N}}$ is not countable. □

1.9 Countability

Recall Definition 1.2.7 of a basis of a topology. A more refined notion is the following:

Definition 1.9.1: Let X be a topological space and $x_0 \in X$. A *neighborhood basis* at x_0 is a family $\{N_\alpha\}_{\alpha \in I}$ of neighborhoods of x_0 such that every neighborhood N of x_0 contains some $N_\alpha \subset N$, $\alpha \in I$.

Remark 1.9.2: The topology is determined by giving a neighborhood basis at every point $x \in X$.

Remark 1.9.3: In a metric space X, $\{B_{1/n}(x_0)\}_{n \in \mathbb{N}}$ is a countable neighborhood basis of $x_0 \in X$.

Definition 1.9.4: A topological space X is called

(i) *first countable* if there is a countable neighborhood basis at every point $x_0 \in X$.

(ii) *second countable* if there is a countable basis of the topology.

Remark 1.9.5: Clearly, second countability implies first countability.

Let X be an uncountable set with the discrete topology $\mathcal{T} = 2^X$ (i.e., every subset of X is open). Then X is first countable, but not second countable. Thus the two notions are inequivalent.

Remark 1.9.6: Remark 1.9.3 shows that a metrizable space X is first countable.

If X is additionally separable, then second countability follows:

Lemma 1.9.7: Let X be a separable metric space. Then there is a countable basis.

Proof. Let $\{x_1, x_2, \ldots\}$ be a countable dense subset of X (i.e., a dense sequence $\{x_j\}_{j \in \mathbb{N}}$). Then

$$\{B_{1/n}(x_j)\}_{n,j \in \mathbb{N}}$$

is a countable family of open subsets. Let $U \subset X$ be an arbitrary open subset. We claim that

$$U = \bigcup_{(n,j):\, B_{1/n}(x_j) \subset U} B_{1/n}(x_j).$$

Indeed, "$\supset$" is obvious. To show the opposite inclusion, let $x \in U$. Then there is some $n > 0$ such that $B_{1/n}(x) \subset U$. Pick some arbitrary $j \in \mathbb{N}$ such that $d(x, x_j) < 1/(2n)$ (exists by denseness of the sequence $\{x_j\}_{j \in \mathbb{N}}$). Then $x \in B_{1/(2n)}(x_j)$ by definition and the triangle inequality implies that

$$d(x, z) \leq d(x, x_j) + d(x_j, z) < 1/n \qquad \text{for all } z \in B_{1/(2n)}(x_j),$$

i.e.,

$$x \in B_{1/(2n)}(x_j) \subset B_{1/n}(x) \subset U.$$

This shows that $x \in \bigcup_{(n,j):\, B_{1/n}(x_j) \subset U} B_{1/n}(x_j)$ as claimed. □

1.10 Continuous maps between topological spaces

Definition 1.10.1: Let (X, d) and (X', d') be metric spaces, $f : X \to X'$ and $x_0 \in X$. Then f is called *continuous at* x_0 if for all $\varepsilon > 0$ there is some $\delta > 0$ such that $d'(f(x), f(x_0)) < \varepsilon$ whenever $d(x, x_0) < \delta$. That is, $f(x) \in B_\varepsilon(f(x_0))$ for all $x \in B_\delta(x_0)$ or $B_\delta(x_0) \subset f^{-1}\big(B_\varepsilon(f(x_0))\big)$.

Lemma 1.10.2: Let (X, d), (X', d') be metric spaces, $f : X \to X'$. Then the following are equivalent:

(i) f is continuous at x_0.

(ii) For every neighborhood N of $f(x_0)$, the set $f^{-1}(N)$ is a neighborhood of x_0.

Proof. This follows immediately from the definitions. □

Definition 1.10.3: Let $(X, \mathcal{T})$ and $(X', \mathcal{T}')$ be topological spaces and $f : X \to X'$. Then f is *continuous at* $x_0 \in X$ if $f^{-1}(N)$ is a neighborhood of x_0 for every neighborhood N of $f(x_0)$. The function f is called *continuous* if f is continuous at every point $x_0 \in X$. In particular, f is continuous if and only if $f^{-1}(U)$ is open for every open $U \subset X'$.

Lemma 1.10.4: Let (X, d) and (X', d') be metric spaces, $f : X \to X'$ and $x \in X$. Then the following are equivalent:

(i) f is continuous at x.

(ii) $f(x_n) \to f(x)$ for $n \to \infty$ for every sequence $(x_n)_{n\in\mathbb{N}}$ with $x_n \to x$ for $n \to \infty$.

Proof. Analysis I. □

Remark 1.10.5: The statement of Lemma 1.10.4 does not hold for general topological spaces: (i) ⇒ (ii) still holds, but (ii) is not sufficient to guarantee continuity.

Definition 1.10.6: Let X, X' be topological spaces and $f : X \to X'$ bijective. Then f is called a *homeomorphism* if f and f^{-1} are continuous.

Remark 1.10.7: By definition, f is a homeomorphism if and only if f is bijective and the following condition holds:

$$f(U) \subset X' \text{ is open} \quad \Longleftrightarrow \quad U \subset X \text{ is open.}$$

The same characterization holds with *open* replaced by *closed*.

The following shows that compactness is a topological property:

Lemma 1.10.8: Let X, Y be topological spaces where X is compact. Let $f : X \to Y$ be continuous. Then $f(X) \subset Y$ is compact.

Proof. Let $\{G_\alpha\}_{\alpha\in I}$ be an open cover of $f(X)$. Then $f^{-1}(G_\alpha)$ is open for every $\alpha \in I$, and $\{f^{-1}(G_\alpha)\}_{\alpha\in I}$ is an open cover of X. This is because for every $x \in X$ there is some $\alpha \in I$ such that $f(x) \in G_\alpha$, which is equivalent to $x \in f^{-1}(G_\alpha)$.

By compactness of X there is a finite subcover $\{f^{-1}(G_\alpha)\}_{\alpha\in J}$ (i.e., $J \subset I$ with $|J| < \infty$). We conclude that $\{G_\alpha\}_{\alpha\in J}$ is a finite open cover of $f(X)$. □

Corollary 1.10.9: Let $X \neq \emptyset$ be compact and let $f : X \to \mathbb{R}$ be continuous. Then the maxima and minima $\max_{x\in X} f(x)$ and $\min_{x\in X} f(x)$ exist.

Proof. The subset $f(X) \subset \mathbb{R}$ is compact, hence bounded and closed according to Heine–Borel. This implies the claim. □